

Project Proposal

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1. What problem am I solving? State it clearly.

Investigating causal drivers of neighborhood opposition (NIMBYism) to housing. Goal: separate stable, **invariant** predictors of resistance from **spurious correlations** driven by environment-specific confounders (transient contexts). Aim: find features predicting opposition consistently across **market regimes** (2007-2025).

2. Why is this problem important? To whom?

Critical for urban planners. Current models overfit to specific eras, failing when market regimes shift. Identifying *invariant* causes enables policy design addressing fundamental concerns rather than symptoms of a moment, ensuring robustness.

3. What is my strategy for solving it?

Using **Invariant Risk Minimization (IRM)** (Arjovsky et al., 2019):

1. **Environments:** Partition data into $\mathcal{E}_{train}$ (e.g. 07-18) and $\mathcal{E}_{test}$ (19-25).
2. **Representation:** Extract features $\Phi(X)$ from property data and protest letters.
3. **Criterion:** Train $f(\Phi(X))$ to minimize loss across **training** environments only: $\min \sum_{e \in \mathcal{E}_{train}} R^e(f) + \lambda \|\nabla R^e\|^2$.

4. Why is this solution good? Where does it fall short?

Good: Moves beyond correlation to causal structure (Peters et al., 2016). Invariance filters spurious associations (e.g. council tenure) for stable drivers.

Shortfall: Assumes stable causal mechanism. If root reasons for NIMBYism structurally change (e.g. traffic $\rightarrow$ class), invariance is too strong.

5. How can I demonstrate that the solution works?

Test for **OOD Generalization**: train on $\mathcal{E}_{train}$ (07-18, West), test on $\mathcal{E}_{test}$ (19-25, East). Successful Invariant Model maintains performance (AUC) on unseen data where baselines degrade.

References

- Arjovsky, M., Bottou, L., Gulrajani, I., & Lopez-Paz, D. (2019). Invariant Risk Minimization. *arXiv:1907.02893*.
- Peters, J., Bühlmann, P., & Meinshausen, N. (2016). Causal inference by using invariant prediction. *JRSS-B*, 78(5).
- Schölkopf, B., et al. (2021). Towards Causal Representation Learning. *Proceedings of the IEEE*, 109(5).